Wavelet-Guided Temporal Photometric Network for Multi-Frame Infrared Small Target Detection

Abstract— Multi-frame infrared small target detection is challenging due to extremely weak target responses, complex background clutter, and the tendency of tiny targets to be suppressed during deep feature extraction and multi-scale fusion. To address these issues, we propose a wavelet-guided temporal photometric network (WTPNet), a prior-guided three-branch cross-attention framework for moving infrared small target detection. Instead of directly stacking consecutive frames, WTPNet decouples multi-frame information into motion-sensitive temporal cues, context-aware temporal representations, and current-frame spatial appearance. Specifically, a temporal context prior is constructed by jointly using temporal discrete cosine transform and Haar wavelet decomposition, where the former captures motion-sensitive photometric variations across frames and the latter separates low-frequency background structures from high-frequency local details. This prior guides the temporal context branch to emphasize target-related variations while suppressing background-dominated responses. In addition, a directional photometric residual enhancement strategy preserves weak shallow responses in the current frame, and a large-context scale-focused fusion module alleviates target dilution during multi-scale feature aggregation. Experiments on two public datasets demonstrate that WTPNet outperforms representative single-frame and multi-frame detectors. Compared with the baseline, WTPNet improves average precision from 59.10% to 64.06% and from 85.85% to 90.11% on the two datasets, respectively. These results validate the complementarity of temporal-frequency prior learning, photometric response preservation, and scale-focused fusion for robust moving infrared small target detection. The code will be released at https://github.com/vazheaven/WTPNet.

Keywords: Infrared small target detection; Multi-frame detection; Wavelet prior; Temporal context; Photometric enhancement

1. Introduction

Infrared small target detection (IRSTD) is a fundamental task in infrared search and tracking systems, space-based surveillance, maritime monitoring, and remote sensing interpretation. Unlike generic object detection in visible imagery, infrared small targets usually occupy only a few pixels and lack stable shape, texture, and semantic cues [1]. Their weak radiometric responses are easily overwhelmed by structured clutter, sensor noise, and background thermal variations, leading to frequent missed detections and false alarms. Recent studies further suggest that, without explicit preservation of target-related high-gradient or high-frequency cues, small-target signatures can be progressively weakened during feature downsampling and deep semantic abstraction [2, 3]. As illustrated in Fig. 1, the compact responses of infrared small targets tend to attenuate and diffuse as the network hierarchy deepens. Therefore, preserving weak yet discriminative target cues remains a central challenge in infrared small target detection.

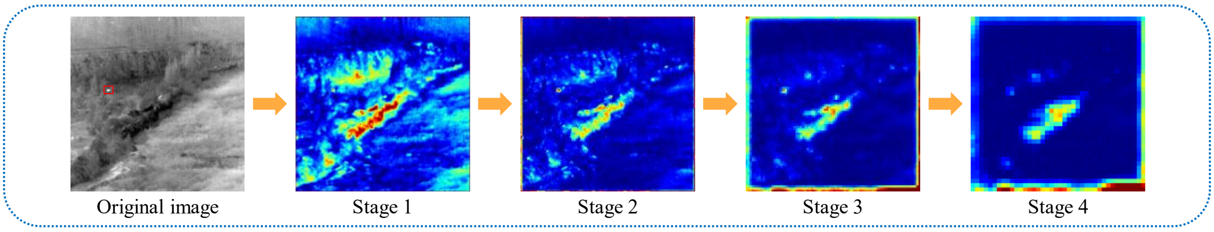


Fig. 1. Progressive attenuation and dilution of infrared small target responses in hierarchical feature learning.

Early IRSTD methods were predominantly model-driven and relied on handcrafted domain priors. Local-contrast approaches enhance targets by exploiting the intensity differences between small targets and their neighboring backgrounds [4, 5]. Low-rank sparse decomposition methods formulate an infrared image or sequence as a superposition of a low-rank background and sparse target components [6, 7], while tensor-based extensions further exploit nonlocal and spatio-temporal correlations for background suppression [8]. For infrared sequences, traditional methods also incorporate temporal profiles and motion saliency to distinguish weak moving targets from complex backgrounds [9, 10]. These approaches are interpretable and have revealed useful priors, such as local photometric contrast, background low-rankness, and temporal consistency. Nevertheless, their performance is often sensitive to manually designed windows, thresholds, rank assumptions, and iterative optimization schemes. More critically, strong edges, isolated noise, and background corners may also exhibit sparse or high-frequency responses, making purely handcrafted priors insufficiently discriminative for separating true targets from clutter-induced pseudo responses in complex infrared scenes.

With the development of deep learning, single-frame infrared small target detection has substantially improved learnable feature representation and background suppression [11]. Existing methods have explored U-shaped encoder–decoder architectures [12], dense nested feature interactions [13], high-resolution feature preservation [14], multi-scale attention mechanisms [15], Transformer-based context modeling [16], and detection-oriented prediction heads [17] to compensate for the weak semantics and extremely small spatial extent of infrared targets. Recent studies further introduce image denoising, wavelet-based downsampling, and spatial–frequency fusion to preserve local details and enhance target–background separability [18, 19]. Despite these advances, most single-frame methods still rely primarily on spatial appearance cues from an individual image. When the target-to-background contrast is extremely low, spatial evidence alone is often insufficient to reliably distinguish true targets from clutter-like responses. In addition, repeated downsampling and deep semantic abstraction may suppress the compact photometric signatures of small targets, while coarse multi-scale fusion can propagate background interference into fine-resolution details.

To overcome the limitations of single-frame detection, multi-frame methods exploit temporal information from infrared sequences. Traditional sequence-based studies have shown that dim targets can form discriminative photometric patterns along the temporal dimension, such as energy accumulation and temporal-profile similarity [20, 21]. Deep multi-frame approaches further learn motion-related representations from adjacent frames through dual-stream spatio-temporal modeling, cross-attention interaction, and multi-scale temporal fusion [22-25]. Recent efforts have also investigated motion-pattern mining [26], state-space modeling [27], and neural spatio-temporal tensor representation [28], suggesting that temporal information should not be treated merely as additional input frames, but rather as a structured guidance signal. Nevertheless, temporal modeling is not automatically beneficial. Indiscriminate frame stacking or unconstrained temporal aggregation may amplify background-induced temporal variations together with true target responses. Therefore, a more discriminative temporal context prior is needed to emphasize target-related temporal changes while suppressing background-dominated responses.

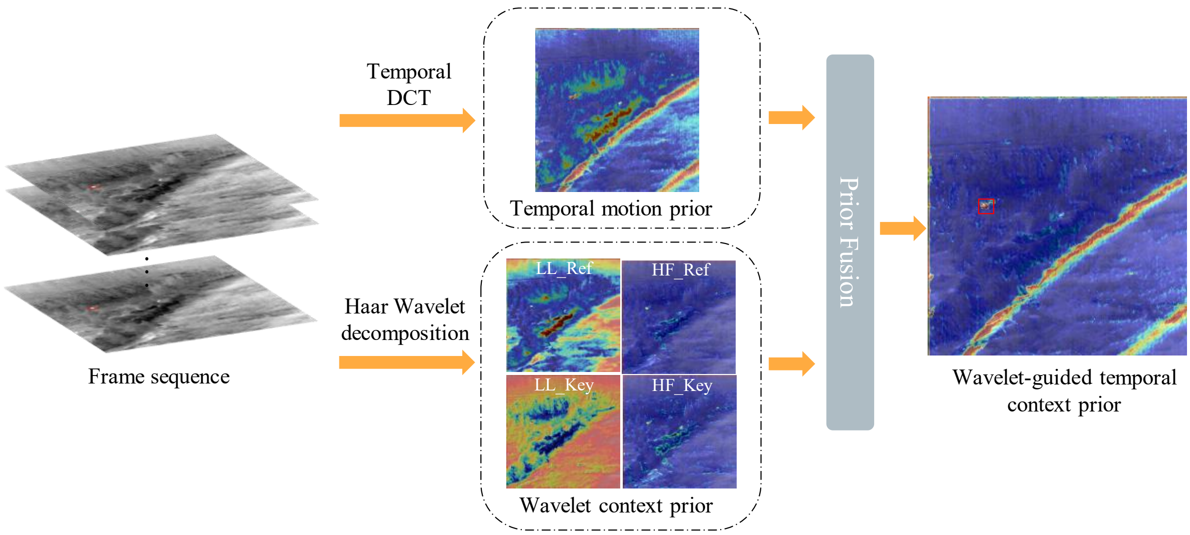


Fig. 2. Motivation of the proposed wavelet-guided temporal context prior.

The above discussion suggests that effective multi-frame infrared small target detection should satisfy three coupled requirements. First, the weak photometric evidence of small targets should be preserved at shallow stages before it is submerged by background structures or weakened by deep semantic abstraction. This is particularly important because true infrared targets and clutter-like structures often exhibit different directional response patterns, making direction-aware local enhancement valuable for early feature learning. Second, temporal information should be converted into target-related contextual guidance rather than aggregated indiscriminately. Frequency and wavelet-guided studies have shown that small targets are often more distinguishable in high-frequency or cross-domain representations, whereas low-frequency components mainly reflect slowly varying background structures [29]. Third, hierarchical feature fusion should avoid diluting compact target responses with large-scale contextual semantics. Although multi-scale and cross-level fusion can combine fine spatial details with deep contextual information, direct fusion may cause semantic mismatch and propagate background interference [30]. These requirements correspond to three complementary aspects of the same problem: preserving weak target responses, discriminating target-related temporal context, and maintaining scale-aware feature focus.

To this end, we propose a wavelet-guided temporal photometric network (WTPNet) for multi-frame IRSTD. WTPNet adopts a three-branch cross-attention architecture that decouples multi-frame representation learning into motion-sensitive temporal cues, context-aware temporal representations, and current-frame spatial appearance. As illustrated in Fig. 2, the proposed wavelet-guided temporal context prior (WTCP) combines temporal discrete cosine transform with Haar wavelet decomposition. The temporal transform captures motion-sensitive photometric variations across frames, while the Haar wavelet decomposition separates low-frequency background structures from high-frequency local details. Their combination provides a more discriminative temporal context prior, enabling the context branch to emphasize target-related temporal variations and suppress background-dominated responses. In parallel, a directional photometric residual enhancement (DPRE) module preserves weak shallow responses in the current-frame branch, and a large-context scale-focused fusion (LCSF) module refocuses hierarchical features on compact target regions under broad contextual perception. In this way, WTPNet forms a unified prior-guided framework that improves multi-frame infrared small target detection from three complementary perspectives: temporal context purification, shallow photometric response preservation, and scale-aware feature fusion.

The main contributions of this paper are summarized as follows:

1) We propose WTPNet, a prior-guided three-branch cross-attention framework for multi-frame infrared small target detection, which jointly models motion-sensitive temporal cues, context-aware temporal representations, and current-frame spatial appearance.
2) We design a wavelet-guided temporal context prior module that combines temporal discrete cosine transform and Haar wavelet decomposition to improve temporal-context discrimination under dynamic infrared clutter.
3) We introduce a directional photometric residual enhancement module and a large-context scale-focused fusion module to preserve weak shallow responses and alleviate target dilution during hierarchical feature fusion, respectively.
4) Extensive experiments on two public datasets show that WTPNet outperforms representative single-frame and multi-frame methods. Compared with the baseline, WTPNet improves AP50 from 59.10% to 64.06% and from 85.85% to 90.11%, respectively.

2. Related Work

2.1 Single-frame infrared small target detection

Single-frame infrared small target detection is one of the fundamental settings in IRSTD. Early deep learning methods usually formulated IRSTD as a dense segmentation problem, where a central challenge is to preserve weak target responses throughout hierarchical feature extraction. DNANet mitigates the disappearance of small targets in deep layers through dense nested feature interaction and channel–spatial attention [13]. For spaceborne thermal infrared tiny ship detection, HFA-Net further introduces multi-level detail enhancement, large-kernel attention, and hierarchical feature fusion to strengthen weak target representations [31]. To improve contextual modeling, IR-TransDet incorporates Transformer-based global attention for infrared dim and small target detection [32], while MPCNet reduces the semantic gap between low-level spatial details and high-level contextual features through multi-scale perception and cross-attention fusion [33].

Recent studies have also moved beyond direct target enhancement and begun to model the target–background relationship more explicitly. PBT progressively aligns candidate target responses with background context, showing that background information can provide useful cues for false-alarm suppression [34]. RPCANet unfolds RPCA-inspired optimization into a learnable network, bridging low-rank sparse priors and deep representation learning while improving interpretability [35]. SP-KAN reformulates IRSTD as sparse nonlinear contextual modulation and adopts Kolmogorov–Arnold networks to enhance target–background discrimination under complex clutter [36]. These methods indicate that single-frame IRSTD has evolved from local target enhancement toward structured modeling of target responses, background context, and nonlinear separability.

In addition to segmentation-oriented approaches, detection-oriented frameworks have recently attracted increasing attention. EFLNet treats IRSTD as a bounding-box detection task and addresses target–background imbalance, small-box regression sensitivity, and semantic-level selection [37]. PConv with scale-based dynamic loss further suggests that both convolutional operators and optimization objectives should be adapted to the Gaussian-like spatial distribution and scale-sensitive localization of infrared small targets [38]. SpecDETR extends point-object detection to hyperspectral imagery, providing a broader instance-level perspective through joint spatial–spectral representation [39]. Despite these advances, single-frame methods remain constrained by the limited spatial evidence available in an individual image, especially when dim targets are submerged by clutter-like responses. This limitation motivates the use of temporal photometric cues from adjacent frames, while still requiring careful preservation of shallow target responses and scale-aware feature fusion.

2.2 Multi-frame infrared small target detection

Multi-frame infrared small target detection has become an important paradigm for recognizing weak targets that are difficult to distinguish from a single frame. General video representation studies have shown that 3D CNNs and ConvLSTMs can jointly encode local spatial correlations and temporal dependencies, providing an early foundation for spatio-temporal feature learning [40]. In infrared small target detection, STDMANet extracts temporal features at multiple time scales and refines them with spatial multi-scale attention [41], while ST-Trans introduces spatial–temporal Transformer modeling to establish inter-frame dependencies in infrared sequences [42]. These methods indicate that temporal context is crucial for detecting extremely weak targets, but the effectiveness of multi-frame IRSTD largely depends on how temporal information is extracted, filtered, and fused.

A major line of research focuses on explicit motion modeling from frame sequences. DTUM encodes motion direction through motion-to-data mapping and direction-coded convolution, making the difference between target trajectories and clutter trajectories more measurable [43]. SSTNet extends spatio-temporal modeling to cross-slice motion representation and exploits past–current–future contexts through cross-slice ConvLSTM [44]. TMP decouples temporal motion perception from spatial auxiliary enhancement and performs cross-domain spatio-temporal fusion via complementary symmetry weighting [45]. More recently, TDCNet integrates temporal difference modeling with 3D convolution to capture motion-context dependencies while maintaining spatio-temporal representation capacity [46].

Another research direction addresses dynamic backgrounds and platform motion, which are common in practical infrared sequences. GST-Det extracts relative motion patterns through optical-flow accumulation and performs global spatio-temporal fusion to improve target–background discrimination in complex ground scenes [47]. MFE-Net further considers spaceborne dynamic scenarios and uses motion-guided feature refinement with multi-attention fusion to suppress dynamic clutter and enhance target responses [48]. LMAFormer aligns multi-frame features with local motion awareness and models motion backgrounds using a multi-scale Transformer [16]. MOCID introduces Fourier-inspired spatio-temporal attention and displacement-aware Mamba to learn clip-level motion context and frame-level displacement information, respectively [49].

2.3 Frequency- and Wavelet-guided Representation Learning

Frequency- and wavelet-guided representation learning has recently attracted increasing attention in infrared small target detection. The underlying motivation is that infrared small targets usually appear as local intensity discontinuities or weak compact bright spots, which can be easily smoothed by convolutional propagation and downsampling in purely spatial-domain networks. HDNet builds a hybrid spatial–frequency framework, where the spatial branch enhances multi-scale target contrast and the frequency branch suppresses slowly varying low-frequency background components [50]. FADet introduces Haar wavelet decomposition into the visual encoder and uses high-frequency sub-bands to generate attention masks, thereby preserving fine-grained target details and mitigating over-smoothing [51]. FGARNet further analyzes infrared small targets from a signal- and frequency-domain perspective, combining frequency filtering with wavelet-rectified downsampling to enhance target-related frequency responses and reduce aliasing-induced feature distortion [52].

Wavelet decomposition has also been explored for feature fusion and noise suppression in IRSTD. WCDMF-Net constructs a parallel spatial–frequency dual-stream architecture, extracting multi-scale frequency cues with Haar wavelet transform and inverse reconstruction before fusing them with spatial context representations [19]. CSAFNet incorporates a wavelet-based dynamic enhancement branch into context-aware feature extraction to strengthen local details under semantic-guided adaptive filtering [53]. Different from methods that simply emphasize high-frequency details, NS-FPN points out that high-frequency components may also introduce false alarms and therefore uses low-frequency information to purify high-frequency features during feature pyramid fusion [54].

Beyond single-frame IRSTD, frequency-domain modeling has been introduced into multi-frame infrared small target detection and broader tiny-object detection tasks. Tridos extends moving infrared small target detection from the conventional spatio-temporal domain to a triple-domain representation, using Fourier-based local–global frequency modeling to enhance spatio-temporal features and residual compensation to reduce cross-domain mismatch [55]. In remote sensing tiny-object detection, SMWG-DETR combines Fourier spectral modulation with wavelet-guided fusion to suppress redundant spectral responses and restore local high-frequency details across scales [29]. In addition, IRPruneDeXt uses wavelet-domain regularization for channel pruning, showing that wavelet priors can also support efficient infrared small target model compression, although its focus is not feature enhancement itself [3].

In summary, existing studies have advanced infrared small target detection from three complementary perspectives: single-frame spatial representation, multi-frame temporal modeling, and frequency- or wavelet-guided feature learning. However, these directions are still not fully integrated for moving infrared small target detection. Single-frame methods are constrained by limited spatial evidence, especially when weak targets are submerged by clutter-like responses. Multi-frame methods exploit temporal cues, but their temporal aggregation may still enhance background-induced fluctuations together with true target responses. Frequency- and wavelet-guided methods preserve fine local details, yet most of them rely on static spatial–frequency priors and do not explicitly purify temporal context. These observations motivate us to develop WTPNet, which introduces a wavelet-guided temporal context prior into a three-branch cross-attention framework, while preserving shallow photometric responses and improving scale-focused feature fusion for robust multi-frame infrared small target detection.

3. Method

3.1 Overall Architecture

The overall architecture of the proposed WTPNet is shown in Fig. 3. Given an infrared frame sequence $X = \{X_i\}_{i=1}^{T}$ and the current frame X_T, WTPNet adopts a three-branch cross-attention framework to jointly exploit motion-sensitive temporal cues, context-aware temporal representations, and spatial appearance information from the current frame. Rather than directly stacking multi-frame features or simply fusing 2D and 3D representations, WTPNet decouples complementary information into query, key, and value branches and conducts cross-attention interaction across multi-scale hierarchies, thereby enhancing spatiotemporal features with stronger discriminability.

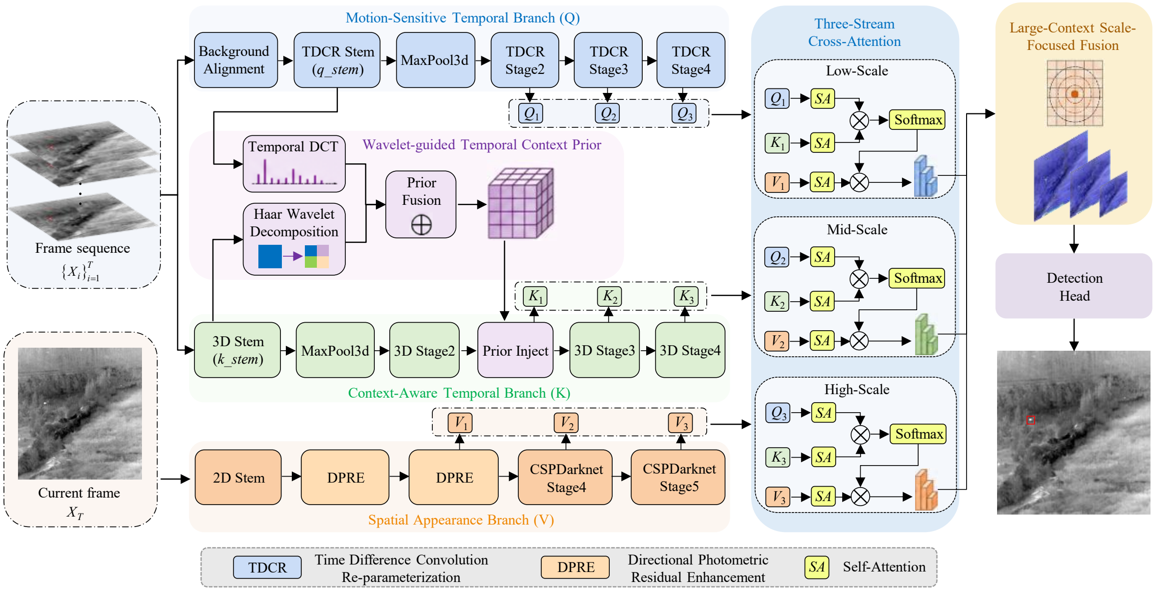


Fig. 3. Overall architecture of the proposed WTPNet. WTPNet consists of a motion-sensitive temporal branch (Q), a context-aware temporal branch (K), and a spatial appearance branch (V). The WTCP module introduces wavelet-guided temporal context priors into the K branch, while DPRE enhances shallow photometric responses in the V branch. Multi-scale cross-attention is performed with Q, K, and V as query, key, and value, respectively. The enhanced features are further fused by LCSF and sent to the detection head.

Specifically, the motion-sensitive temporal branch Q processes the aligned frame sequence with a TDCR-based backbone [46] and produces multi-scale motion features $\{Q_1, Q_2, Q_3\}$. The context-aware temporal branch K extracts 3D spatio-temporal features $\{K_1, K_2, K_3\}$ from the same sequence, where the proposed WTCP module is introduced to provide wavelet-guided temporal context priors. The spatial appearance branch V takes only the current frame X_T as input and employs DPRE-enhanced CSPDarknet stages to preserve weak photometric responses of infrared small targets. The resulting multi-scale spatial features are denoted as $\{V_1, V_2, V_3\}$. For each scale i, the three branches are first refined by self-attention and then integrated through cross-attention, where Q_i, K_i and V_i serve as query, key, and value, respectively. This design uses motion-sensitive temporal features to query target-relevant regions, wavelet-prior-enhanced temporal features to provide contextual matching cues, and current-frame spatial features to supply precise photometric details. The enhanced multi-scale features are then aggregated by the proposed LCSF module and forwarded to the detection head for final target localization.

Overall, WTPNet forms a unified prior-guided detection framework. WTCP improves temporal context discrimination, DPRE strengthens shallow spatial photometric responses, and LCSF mitigates target dilution during hierarchical feature fusion. These components complement the three-branch cross-attention design and jointly improve the robustness of multi-frame infrared small target detection.

3.2 Wavelet-Guided Temporal Context Prior

Although 3D convolution can encode local spatio-temporal patterns, it treats target-related variations and background-induced fluctuations in a largely uniform manner. In infrared sequences, weak small targets usually appear as compact photometric changes, whereas clouds, edges, sensor noise, and thermal clutter often dominate the temporal response. Directly using such raw temporal features as the key stream in cross-attention may therefore introduce ambiguous contextual cues. To address this issue, we propose a wavelet-guided temporal context prior (WTCP) module, which extracts target-related temporal and wavelet-domain priors from early-stage features and injects them into the context-aware temporal branch.

As shown in Fig. 4, WTCP takes the stem features from the motion-sensitive branch (Q) and the context-aware branch (K) as input:

$$F_q \in \mathbb{R}^{B\times C_q\times T\times H\times W}, \qquad F_k \in \mathbb{R}^{B\times C_k\times T\times H\times W}. \tag{1}$$

The two features are first projected into compact representations by lightweight convolutional layers:

$$Z_q = \phi_q(F_q), \qquad Z_k = \phi_k(F_k). \tag{2}$$

where $\phi_q(\cdot)$ and $\phi_k(\cdot)$ denote $1\times1\times1$ Conv-BN-SiLU projections. WTCP then constructs two complementary priors, namely a DCT-based motion prior P_q and a wavelet-based context prior P_k.

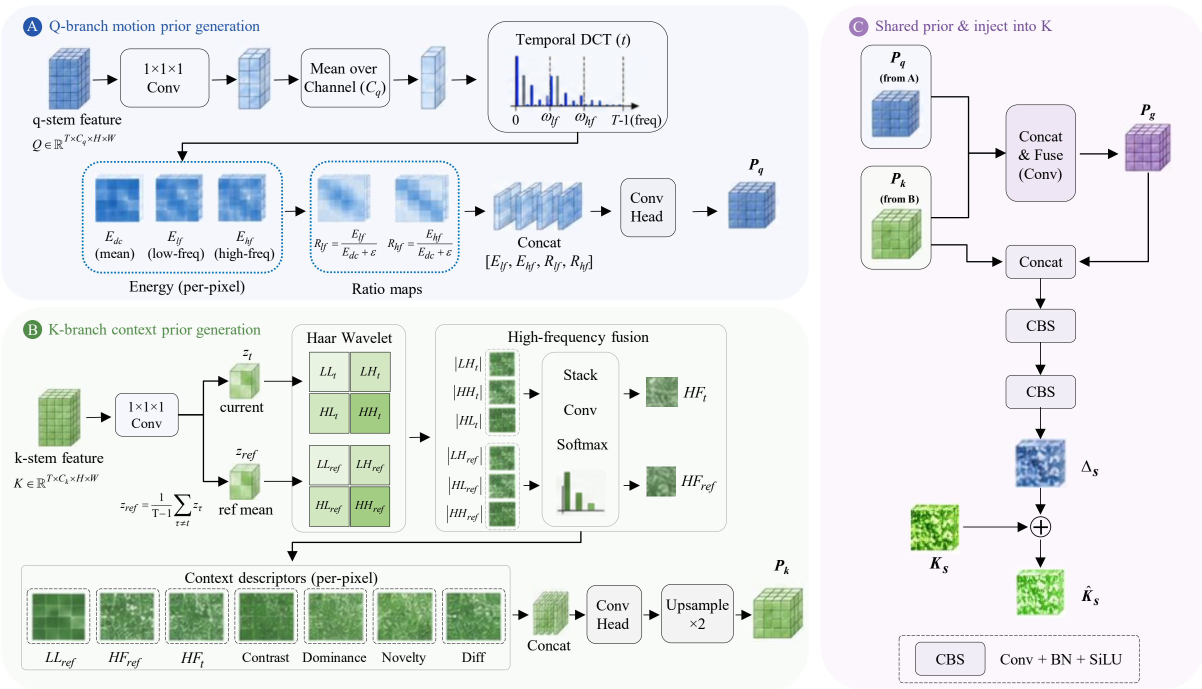


Fig. 4. Structure of the proposed wavelet-guided temporal context prior module. WTCP first extracts a DCT-based motion prior from the Q-branch stem feature and a Haar wavelet-based context prior from the K-branch stem feature. The two priors are fused into a shared prior and injected into the K-branch through a residual compensation scheme, guiding the context-aware temporal features toward target-related temporal and wavelet-domain responses.

For the Q-branch, we summarize $\bar{Z}_q$ along the channel dimension and apply temporal DCT at each spatial location. Let $\mathcal{D}t(\cdot)$ denote the temporal DCT operator. The temporal coefficients are obtained by $A = \mathcal{D}t(\bar{Z}q) = \{A_0, A_1, ..., A_{T-1}\}$, where $\bar{Z}_q \in \mathbb{R}^{B\times T\times H\times W}$ is the channel-averaged response. We divide the DCT coefficients into DC, low-frequency, and high-frequency components, and compute their energy maps as:

$$E_{dc} = A_0^2,$$
$$E_{lf} = \frac{1}{|\Omega_l|}\sum_{f\in\Omega_l} A_f^2, \quad \Omega_l = \{1, 2, \ldots, \min(2, T-1)\}, \tag{3}$$
$$E_{hf} = \frac{1}{|\Omega_h|}\sum_{f\in\Omega_h} A_f^2, \quad \Omega_h = \{3, 4, \ldots, T-1\}.$$

where Ω_l and Ω_h denote the low- and high-frequency temporal bands, respectively. To describe relative temporal changes with respect to the dominant response, we further compute:

$$R_{lf} = \frac{E_{lf}}{E_{dc} + \epsilon}, \qquad R_{hf} = \frac{E_{hf}}{E_{dc} + \epsilon}. \tag{4}$$

where ϵ is a small positive constant for numerical stability, which prevents the relative frequency responses from being dominated by near-zero DC energy. The motion prior is then generated as $P_q = \mathcal{H}q\left(\mathrm{Concat}[E_{lf}, E_{hf}, R_{lf}, R_{hf}]\right)$, where $\mathcal{H}q(\cdot)$ is a lightweight convolutional head. This prior emphasizes local regions whose temporal energy distribution is more consistent with target-related motion-sensitive changes.

For the K-branch, WTCP further constructs a wavelet-domain context prior. We use the last temporal slice as the current feature and the mean of previous slices as the reference context:

$$Z_T = Z_{k,T}, \qquad Z_{ref} = \frac{1}{T-1}\sum_{t=1}^{T-1} Z_{k,t}. \tag{5}$$

The current-reference difference is defined as $D_T = |Z_T - Z_{ref}|$. A 2D Haar wavelet transform $\mathcal{W}(\cdot)$ is then applied to both Z_T and Z_{ref}:

$$(LL_u, LH_u, HL_u, HH_u) = \mathcal{W}(Z_u), \qquad u \in \{T, ref\}. \tag{6}$$

Here, LL_u represents the low-frequency structural component, while LH_u, HL_u, and HH_u describe directional high-frequency details. Since different high-frequency sub-bands contribute unequally under different clutter patterns, we adaptively fuse them into a compact high-frequency response:

$$HF_u = \sum_{r\in LH, HL, HH} \omega_u^r |r_u|, \qquad u \in \{T, ref\}. \tag{7}$$

where the sub-band weight ω_u^r is obtained by a channel-wise softmax over the three high-frequency responses. Based on the wavelet decomposition, we construct a set of context descriptors:

$$\mathcal{C}_k = \mathrm{Concat}[LL_{ref}, HF_{ref}, HF_T, C_T, B_{ref}, N_T, \Delta_T]. \tag{8}$$

Here, $C_T = \log\left(1 + \frac{HF_T}{|LL_T| + \epsilon}\right)$ measures the high-to-low frequency contrast of the current feature. $B_{ref} = \log\left(1 + \frac{|LL_{ref}|}{HF_{ref} + \epsilon}\right)$ describes the low-frequency dominance of the reference background. $N_T = \log\left(1 + \frac{|HF_T - HF_{ref}|}{HF_{ref} + \epsilon}\right)$ captures the high-frequency novelty of the current feature relative to the reference context. In addition, $\Delta_T = \mathcal{P}(D_T)$ encodes the current-reference spatial variation at the wavelet scale, where $\mathcal{P}(\cdot)$ denotes spatial pooling. The wavelet-based context prior is produced by $P_k = \mathrm{Up}\big(\mathcal{H}_k(\mathcal{C}_k)\big)$, where $\mathcal{H}_k(\cdot)$ is a lightweight convolutional head and $\mathrm{Up}(\cdot)$ restores the prior to the stem resolution.

The motion prior P_q and the wavelet context prior P_k are complementary: the former captures temporal energy redistribution, while the latter separates low-frequency background structure from high-frequency target-related details. We therefore fuse them into a shared prior: $P_g = \mathcal{H}_g([P_q, P_k])$, where $\mathcal{H}_g(\cdot)$ denotes a 1×1 Conv-BN-SiLU fusion layer.

Finally, WTCP injects the generated priors into the selected stages of the *K*-branch through residual compensation. Specifically, the context prior P_k and the shared prior P_g are first transformed into a multi-scale prior pyramid, producing stage-wise priors P_k^s and P_g^s that match the spatial resolution of the s-th *K*-branch feature K_s. The prior-induced residual is then computed as

$$\Delta_s = \mathcal{M}_s([P_k^s, P_g^s]). \tag{9}$$

where $\mathcal{M}_s(\cdot)$ denotes a lightweight convolutional mapping that projects the concatenated priors to the channel dimension of K_s. The refined key feature is obtained by

$$\hat{K}_s = K_s + \lambda_s \Delta_s, \tag{10}$$

where λ_s is a learnable scaling factor. This residual formulation allows WTCP to introduce target-related temporal and wavelet-domain guidance without overwriting the original 3D spatio-temporal representation of the *K*-branch.

Through WTCP, the context-aware temporal branch is no longer guided only by raw 3D spatio-temporal features. Instead, it is calibrated by a compact prior that jointly models temporal energy variation, wavelet-domain background structure, and high-frequency target novelty. Consequently, the key features used in subsequent three-stream cross-attention become more discriminative, enabling WTPNet to better suppress background-dominated temporal responses and enhance weak infrared small targets.

3.3 Directional Photometric Residual Enhancement

In the proposed three-stream cross-attention framework, the spatial appearance branch *V* serves as the value stream and provides the current-frame photometric evidence for final target localization. Therefore, even when temporal motion and context features offer effective guidance, the detector still depends on whether the current-frame branch can preserve sufficiently discriminative local responses. This is non-trivial for infrared small targets, because their compact photometric responses are easily smoothed or weakened during early downsampling and semantic abstraction. Meanwhile, many background structures, such as cloud boundaries, tree edges, and sensor-induced streaks, exhibit stronger directional continuity than true targets. This motivates us to enhance shallow spatial features with direction-aware photometric compensation before they are used as value features in cross-attention.

Inspired by the directional asymmetric sampling strategy of PConv [38], we design a directional photometric residual enhancement (DPRE) module for the spatial appearance branch. Different from directly replacing standard convolutions with PConv, DPRE embeds a residual directional enhancer after selected shallow CSPDarknet stages. This design adapts the directional sampling prior to our three-stream architecture: the original CSPDarknet representation is retained as the main appearance feature, while the directional PConv response is used as a mild residual compensation for weak target-related photometric cues.

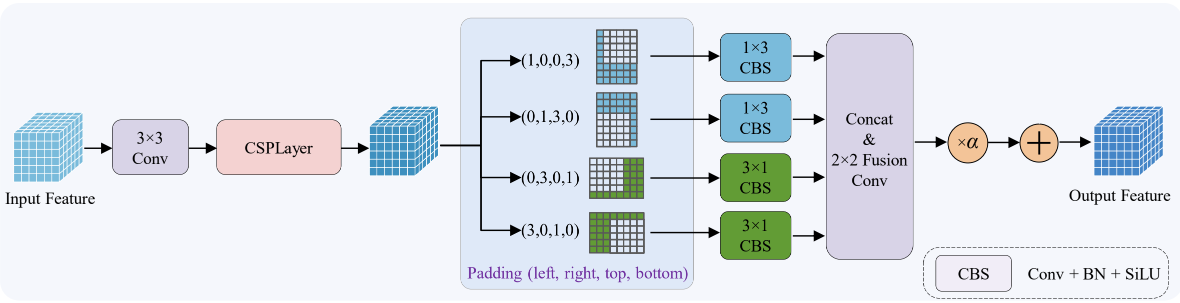


Fig. 5. Structure of the proposed directional photometric residual enhancement module.

Let $F_s \in \mathbb{R}^{B\times C_s\times H_s\times W_s}$ denote the output feature of the s -th CSPDarknet stage. DPRE enhances F_s by

$$\hat{F}_s = F_s + \alpha \cdot \mathcal{P}dir(F_s), \quad s \in \mathcal{S}_e. \tag{11}$$

where $\mathcal{P}dir(\cdot)$ denotes the directional photometric enhancement operator, α is the residual enhancement coefficient, and $\mathcal{S}e$ is the set of enhanced stages. In our implementation, DPRE is applied to the shallow stages ($\mathcal{S}_e = \mathrm{dark2}, \mathrm{dark3}$), and α is set to 0.1. The small residual coefficient prevents the enhancer from over-modifying the original CSPDarknet feature distribution while still introducing directional photometric compensation. The operator $\mathcal{P}_{dir}(\cdot)$ consists of four asymmetric directional branches, as illustrated in Fig. 5. Given F_s, four padded features are first generated by asymmetric padding:

$$F_s^{(i)} = \mathrm{Pad}_{p_i}(F_s), \quad i = 1,2,3,4. \tag{12}$$

where the padding configurations follow the order of left, right, top, and bottom:

$$p_1 = (1,0,0,3), \quad p_2 = (0,1,3,0), \quad p_3 = (0,3,0,1), \quad p_4 = (3,0,1,0).$$

The first two branches use horizontal strip convolution, while the last two branches use vertical strip convolution:

$$U_1 = \mathrm{CBS}_{1\times 3}(F_s^{(1)}), \quad U_2 = \mathrm{CBS}_{1\times 3}(F_s^{(2)}).$$
$$U_3 = \mathrm{CBS}_{3\times 1}(F_s^{(3)}), \quad U_4 = \mathrm{CBS}_{3\times 1}(F_s^{(4)}). \tag{13}$$

where $\mathrm{CBS}_{k_h \times k_w}(\cdot)$ denotes a convolutional block composed of convolution, batch normalization, and SiLU activation. The asymmetric padding shifts the effective sampling regions toward different directions, while the 1×3 and 3×1 strip kernels capture horizontal and vertical photometric variations, respectively. As a result, the four branches form a pinwheel-like directional receptive pattern around the local response.

The directional responses are concatenated and fused to obtain the residual photometric enhancement:

$$\mathcal{P}dir(F_s) = \mathrm{CBS}_{fuse}\left(\mathrm{Concat}[U_1, U_2, U_3, U_4]\right). \tag{14}$$

where $\mathrm{CBS}_{fuse}(\cdot)$ denotes a lightweight fusion block. Therefore, the complete DPRE operation can be written as

$$\hat{F}_s = F_s + \alpha \bullet \mathrm{CBS}_{fuse}\left(\mathrm{Concat}[U_1, U_2, U_3, U_4]\right). \tag{15}$$

Compared with the original use of PConv as a convolution replacement, DPRE uses the directional sampling response as a residual enhancement term. This formulation is more suitable for WTPNet because the spatial branch must provide stable value features for cross-attention. The main CSPDarknet path maintains general spatial representation, while the residual directional path selectively strengthens local photometric responses that are easily suppressed in shallow feature learning. The enhanced features are then used to form the multi-scale value stream, providing more reliable current-frame appearance evidence for the subsequent three-stream cross-attention.

3.4 Large-Context Scale-Focused Fusion

After three-stream cross-attention, WTPNet obtains multi-scale enhanced features that encode motion-sensitive temporal cues, wavelet-guided temporal context, and current-frame spatial appearance. However, directly fusing these features in a conventional top-down manner is not optimal for infrared small target detection. On the one hand, high-level features contain broader semantic context that helps suppress structured background clutter. On the other hand, small targets occupy only a few pixels and are easily diluted when high-level features are repeatedly upsampled and merged with shallow features. Therefore, the fusion module should not only introduce large-context perception, but also retain scale-sensitive local aggregation around small target responses. To this end, we propose a large-context scale-focused fusion (LCSF) module, as shown in Fig. 6. LCSF is inspired by the "see large, focus small" principle of LSNet [56], but is redesigned for multi-scale infrared target feature fusion rather than generic backbone token mixing. Specifically, we use a top-down fusion pathway to progressively merge cross-attended features, and introduce LSBlock after selected fusion stages to perform large-context-guided local aggregation. In this way, LCSF exploits broad contextual perception to distinguish target responses from clutter, while adaptively focusing small-kernel aggregation on local regions where small targets are more likely to appear.

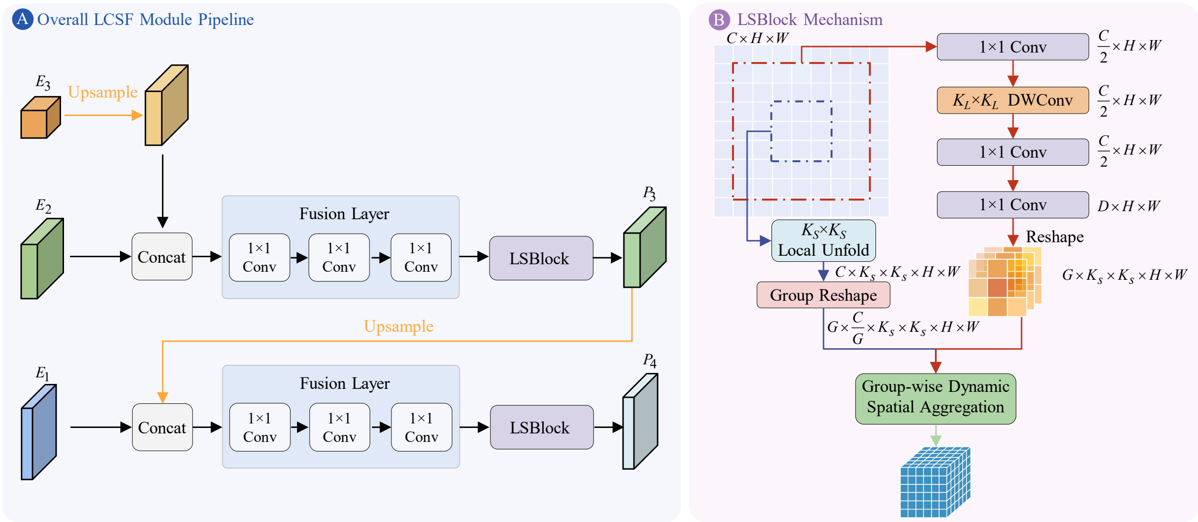


Fig. 6. Structure of the proposed large-context scale-focused fusion module.

Let the three cross-attended features be denoted as E_1, E_2, and E_3, corresponding to low-, mid-, and high-level scales, respectively. The top-down fusion first upsamples the high-level feature and merges it with the mid-level feature:

$$\hat{P}_4 = \mathcal{F}_4\left(\mathrm{Concat}\left[\mathrm{Up}(E_3), E_2\right]\right). \tag{16}$$

where $\mathrm{Up}(\cdot)$ denotes nearest-neighbor upsampling, and $\mathcal{F}_4(\cdot)$ is a lightweight fusion layer composed of consecutive 1×1 convolutional blocks. Then, an LSBlock is used to refine the fused feature:

$$P_4 = \mathrm{LSBlock}(\hat{P}_4). \tag{17}$$

The refined feature P_4 is further upsampled and fused with the low-level feature:

$$\hat{P}_3 = \mathcal{F}_3\left(\mathrm{Concat}\left[\mathrm{Up}(P_4), E_1\right]\right), \tag{18}$$

followed by another LSBlock:

$$P_3 = \mathrm{LSBlock}(\hat{P}_3). \tag{19}$$

The final feature P_3 is sent to the detection head. Compared with standard feature pyramid fusion, this design performs scale-focused refinement after each major fusion step, reducing the risk that weak target responses are overwhelmed by high-level contextual semantics. The core component of LCSF is LSBlock. Given an input feature $X \in \mathbb{R}^{B\times C\times H\times W}$, LSBlock first uses a large-kernel perception (LKP) unit to perceive broad contextual information and generate spatially adaptive aggregation weights. The LKP process is formulated as

$$W = \mathcal{R}\Big(\mathrm{GN}\Big(\mathrm{Conv}1\times1\Big(\mathrm{Conv}1\times1\Big(\mathrm{DWConv}_{K_L\times K_L}\big(\mathrm{Conv}1\times1(X)\big)\Big)\Big)\Big)\Big). \tag{20}$$

where $\mathrm{DWConv}_{K_L\times K_L}(\cdot)$ denotes a large-kernel depth-wise convolution, $\mathrm{GN}(\cdot)$ denotes group normalization, and $\mathcal{R}(\cdot)$ reshapes the generated weights into

$$W \in \mathbb{R}^{B\times G\times K_S^2\times H\times W}. \tag{21}$$

Here, K_L is the large perception kernel size, K_S is the small aggregation kernel size, and G is the number of dynamic groups. In our implementation, $K_L = 7$ and $K_S = 3$, which allows the block to perceive a relatively large neighborhood while restricting feature aggregation to a compact local region.

After large-kernel perception, LSBlock performs small-kernel aggregation (SKA). The input feature X is unfolded into local $K_S \times K_S$ neighborhoods:

$$\mathcal{U}(X) \in \mathbb{R}^{B\times G\times \frac{C}{G}\times K_S^2\times H\times W}, \tag{22}$$

where $\mathcal{U}(\cdot)$ denotes local unfolding and $\frac{C}{G}$ channels share one dynamic aggregation kernel within each group. The dynamic weights are normalized along the kernel dimension:

$$\tilde{W}_{g,k,h,w} = \frac{\exp(W_{g,k,h,w})}{\sum_{j=1}^{K_S^2}\exp(W_{g,j,h,w})}. \tag{23}$$

For the g-th group, the aggregated output is computed as

$$Y_g(:,h,w) = \sum_{k=1}^{K_S^2} \tilde{W}_{g,k,h,w}, \mathcal{U}_k(X_g)(:,h,w), \tag{24}$$

where X_g denotes the feature channels belonging to the g-th group, and $\mathcal{U}_k(X_g)$ denotes the k-th local unfolded element. The outputs of all groups are concatenated to obtain Y. Finally, LSBlock adopts a residual formulation:

$$\mathrm{LSBlock}(X) = X + \mathrm{BN}(Y). \tag{25}$$

This mechanism differs from a simple combination of large-kernel and small-kernel convolutions. The large-kernel branch does not directly replace local aggregation. Instead, it predicts position-adaptive small-kernel weights from broad contextual observations. Thus, the module can use large surrounding context to estimate where reliable local aggregation should be focused, which is particularly suitable for infrared small target detection: background clutter often requires broader context for discrimination, while the target response itself must be preserved through compact local aggregation.

Overall, LCSF integrates three-stream cross-attended features through a top-down pathway and refines the fused representations with large-context-guided small-kernel aggregation. The module complements WTCP and DPRE from the fusion perspective: WTCP improves temporal key discrimination, DPRE preserves current-frame shallow photometric evidence, and LCSF prevents small target responses from being diluted during hierarchical feature fusion.

4. Experiments

4.1 Implementation Details

Datasets. Following the standard evaluation protocol in recent moving infrared small target detection studies, we conduct experiments on two public benchmarks, namely IRDST-H [57] and DAUB-R [44]. These datasets provide complementary evaluation scenarios for dynamic infrared target detection. IRDST-H contains heterogeneous infrared sequences with variable native resolutions and multiple aerial target categories, thus presenting a challenging setting with diverse target appearances and scene conditions. DAUB-R, in contrast, consists of fixed-resolution aircraft sequences, which provides a relatively controlled benchmark for evaluating detection robustness on small moving infrared targets. The official training, validation, and testing partitions are adopted for both datasets to ensure fair and reproducible comparisons with existing methods. The detailed dataset statistics are reported in Table 1.

Table 1. Summary of benchmark properties and official evaluation partitions for public moving infrared small target detection datasets.

Dataset	Sequences	Image size	Target type	Training frames	Validation frames	Test frames
IRDST-H	44	720×480, 992×742, 934×696	helicopter, airplane, drone	8725	2694	5354
DAUB-R	17	256×256	aircraft	7500	2000	4277

Experimental Settings. All experiments are implemented in PyTorch and conducted on a workstation equipped with Ubuntu 20.04, an Intel i9-13900K CPU, and a single NVIDIA RTX 4090 GPU. For a fair comparison, all compared methods are trained and evaluated under the same data partitions and input protocol. Every input frame is resized to 640×640 by letterbox preprocessing to preserve the original aspect ratio. The models are trained for 100 epochs with a batch size of 4. SGD is adopted as the optimizer with an initial learning rate of 1×10^{-2}, a momentum of 0.937, and a weight decay factor of 1×10^{-4}. To improve reproducibility, the random seed is fixed to 2026 during training. During inference, the same letterbox preprocessing is applied, and the confidence threshold and non-maximum suppression (NMS) threshold are set to 0.001 and 0.5, respectively. For segmentation-oriented comparison methods, we retain their original feature extraction structures and adopt a unified detection head [58] to generate bounding-box predictions under the same evaluation protocol.

Evaluation metrics. To quantitatively evaluate detection performance, we adopt four commonly used bounding-box detection metrics, including Precision, Recall, F1 score, and AP50. A detected bounding box is regarded as a true positive only when it is successfully matched with a ground-truth target under the predefined intersection-over-union (IoU) criterion. Otherwise, it is counted as a false positive. Ground-truth targets that are not matched by any prediction are regarded as false negatives. Let TP, FP, and FN denote the numbers of true positives, false positives, and false negatives, respectively. Precision measures the reliability of predicted detections, while Recall reflects the ability of a model to retrieve all existing targets. They are defined as

$$Precision = \frac{TP}{TP+FP},\ Recall = \frac{TP}{TP+FN}. \tag{26}$$

Since Precision and Recall emphasize different aspects of detection quality, we further report the F1 score, which provides a balanced measure by computing their harmonic mean:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}. \tag{27}$$

In addition, AP50 is used to evaluate the overall detection quality at an IoU threshold of 0.5. Unlike Precision, Recall, and F1, which describe the detection performance at a specific operating point, AP50 summarizes the precision–recall behavior over different confidence thresholds. Therefore, it jointly reflects the localization accuracy, confidence ranking quality, and detection stability of a model.

4.2 Comparison with SOTA

4.2.1 Quantitative Comparisons

Table 2 reports the quantitative comparison between WTPNet and 13 representative state-of-the-art methods on IRDST-H and DAUB-R. The compared methods cover both single-frame and multi-frame paradigms. Specifically, the single-frame group includes segmentation-oriented infrared small target detectors, such as ISNet [59], UIUNet [12], DNANet [13], MSHNet [60], and RPCANet [35], as well as recent YOLO-style detection baselines, including YOLOv12-L [61], Hyper-YOLO-M [62], and PConv [38]. The multi-frame group contains representative temporal detection methods, including SSTNet [44] and STMENet [63], two ResUNet-based methods augmented with temporal modeling modules, namely ResUNet_RFR [64] and ResUNet_DTUM [43], and the recent TDCNet [46]. This comparison therefore evaluates not only the absolute detection accuracy of WTPNet, but also its effectiveness against different modeling paradigms.

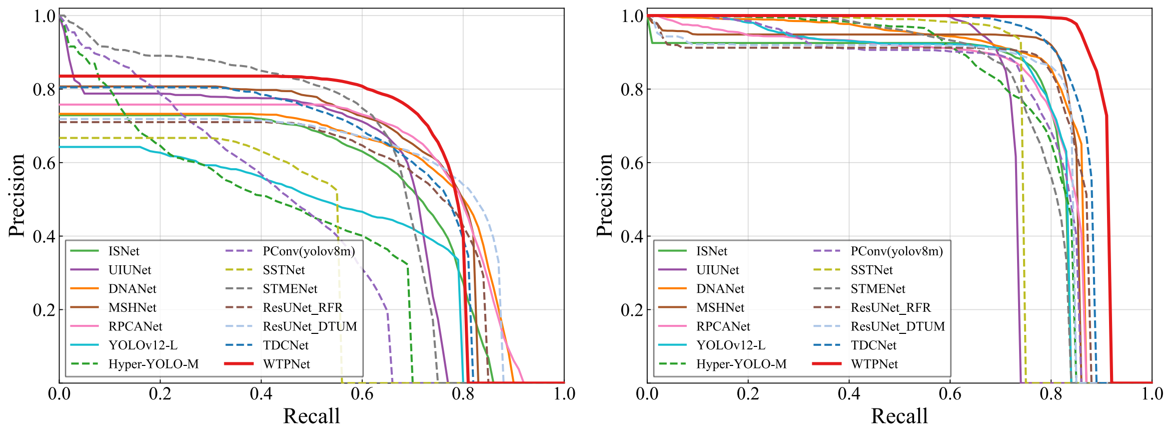
On IRDST-H, WTPNet achieves the best F1 score and AP50 among all compared methods, reaching 80.46% and 64.06%, respectively. Compared with the strongest competing AP50 result, namely MSHNet with 61.44%, WTPNet improves AP50 by 2.62 percentage points. In terms of F1 score, WTPNet also surpasses the second-best method MSHNet by 1.91 percentage points. Although RPCANet obtains a higher recall of 91.09%, its precision drops to 67.59%, indicating that many additional responses are likely introduced. In contrast, WTPNet maintains a more balanced precision–recall trade-off, with 80.36% precision and 80.55% recall, leading to the highest F1 score. This suggests that the proposed method does not simply increase sensitivity to small targets, but improves target discrimination under hard infrared scenarios.

On DAUB-R, WTPNet shows a more pronounced advantage and achieves the best performance across all four metrics. It obtains 99.22% precision, 91.12% recall, 94.99% F1 score, and 90.11% AP50. Compared with TDCNet, the strongest competing multi-frame detector on this dataset, WTPNet improves AP50 from 85.85% to 90.11% and F1 score from 93.08% to 94.99%, corresponding to gains of 4.26 and 1.91 percentage points, respectively. It is also worth noting that WTPNet outperforms the best single-frame baseline by a large margin in AP50, improving over DNANet from 81.57% to 90.11%. These results demonstrate that effectively exploiting temporal cues is particularly beneficial for DAUB-R, where target motion and inter-frame consistency provide complementary information beyond single-frame appearance.

Table 2. Quantitative detection performance comparisons on IRDST-H and DAUB-R test sets. Legend: P: Precision, R: Recall, F1: F1 score, and AP50: average precision at an IoU threshold of 0.5. The best and second-best results are marked in bold and underlined, respectively. All multi-frame methods use five frames.

Type	Methods	Publication	IRDST-H				DAUB-R			
			P	R	F1	AP50	P	R	F1	AP50
Single-frame	ISNet	CVPR 2022	63.54	85.57	72.83	53.54	90.95	83.89	87.28	75.23
	UIUNet	TIP 2022	75.18	74.39	74.78	58.72	98.81	73.37	84.21	72.03
	DNANet	TIP 2023	66.16	<u>89.68</u>	76.14	58.38	95.04	86.02	90.30	81.57
	MSHNet	CVPR 2024	75.18	82.23	<u>78.55</u>	<u>61.44</u>	94.38	85.62	89.79	80.06
	RPCANet	WACV 2024	67.59	**91.09**	77.60	60.93	90.62	86.95	88.75	77.57
	YOLOv12-L	NeurIPS 2025	54.34	79.93	64.63	42.84	93.25	83.91	88.34	77.25
	Hyper-YOLO-M	TPAMI 2025	58.21	69.80	63.48	40.08	92.60	84.87	88.57	77.46
	PConv(yolov8m)	AAAI 2025	64.39	65.87	65.12	41.65	90.71	85.81	88.19	76.74
Multi-frame	SSTNet	TGRS 2024	64.24	55.55	59.58	35.50	<u>99.11</u>	74.37	84.98	73.54
	STMENet	EAAI 2025	**81.53**	74.84	78.04	59.91	93.15	83.70	88.18	76.89
	ResUNet_RFR	TGRS 2025	65.61	84.52	73.87	54.85	89.85	87.61	88.71	77.90
	ResUNet_DTUM	TNNLS 2025	67.28	87.95	76.23	58.28	91.51	84.66	87.95	76.84
	TDCNet	AAAI 2026	73.24	81.77	77.27	59.10	97.90	<u>88.71</u>	<u>93.08</u>	<u>85.85</u>
	WTPNet (Ours)	—	<u>80.36</u>	80.55	**80.46**	**64.06**	**99.22**	**91.12**	**94.99**	**90.11**

Overall, the quantitative results verify the effectiveness of WTPNet for moving infrared small target detection. The consistent improvements in AP50 and F1 score indicate that the proposed method can better integrate temporal cues while preserving discriminative small-target responses, thereby achieving a more favorable balance between accurate localization, target sensitivity, and false-alarm suppression.



(a) (b)

Fig. 7. Precision–recall (PR) curve comparisons of the 14 evaluated methods on the test sets of two benchmark datasets: (a) IRDST-H and (b) DAUB-R.

4.2.2 PR Curve Comparisons

The PR curves in Fig. 7 compare the overall precision–recall behavior of all methods on IRDST-H and DAUB-R. Overall, WTPNet consistently achieves the most favorable precision–recall trade-off on both datasets, with its curve located closest to the upper-right corner. On IRDST-H, where infrared small targets are more easily disturbed by background clutter, most methods suffer from a rapid precision drop as recall increases, whereas WTPNet maintains higher precision over a broader recall range, indicating stronger false-alarm suppression under high-sensitivity detection. On DAUB-R, although several methods obtain high precision at low recall levels, WTPNet preserves near-perfect precision up to a much higher recall region and exhibits delayed performance degradation. These observations are consistent with the AP50 and F1 results in Table 2, further demonstrating that WTPNet improves not only detection accuracy at a fixed threshold but also the overall robustness across different confidence thresholds.

4.2.3 Visualization Comparisons

To provide an intuitive assessment of detection quality, Fig. 8 and Fig. 9 present representative visualization results on IRDST-H and DAUB-R, respectively. The compared methods are evaluated under challenging scenes with dim targets, low target-to-background contrast, and structured background interference. As shown in Fig. 8, several methods tend to miss weak targets around high-contrast background structures on IRDST-H, indicating insufficient discrimination between true small targets and background responses. In contrast, WTPNet produces more complete and cleaner detections, with fewer missed targets and false positives in the enlarged regions. Similar observations can be made on DAUB-R in Fig. 9, where complex textures and background fluctuations frequently cause competing methods to confuse clutter-like responses with true targets. WTPNet still localizes most targets accurately while suppressing spurious detections. These qualitative results are consistent with the quantitative results in Table 2 and the PR curves in Fig. 7, further demonstrating the robustness of WTPNet in maintaining target sensitivity and false-alarm suppression across different infrared moving small-target scenarios.

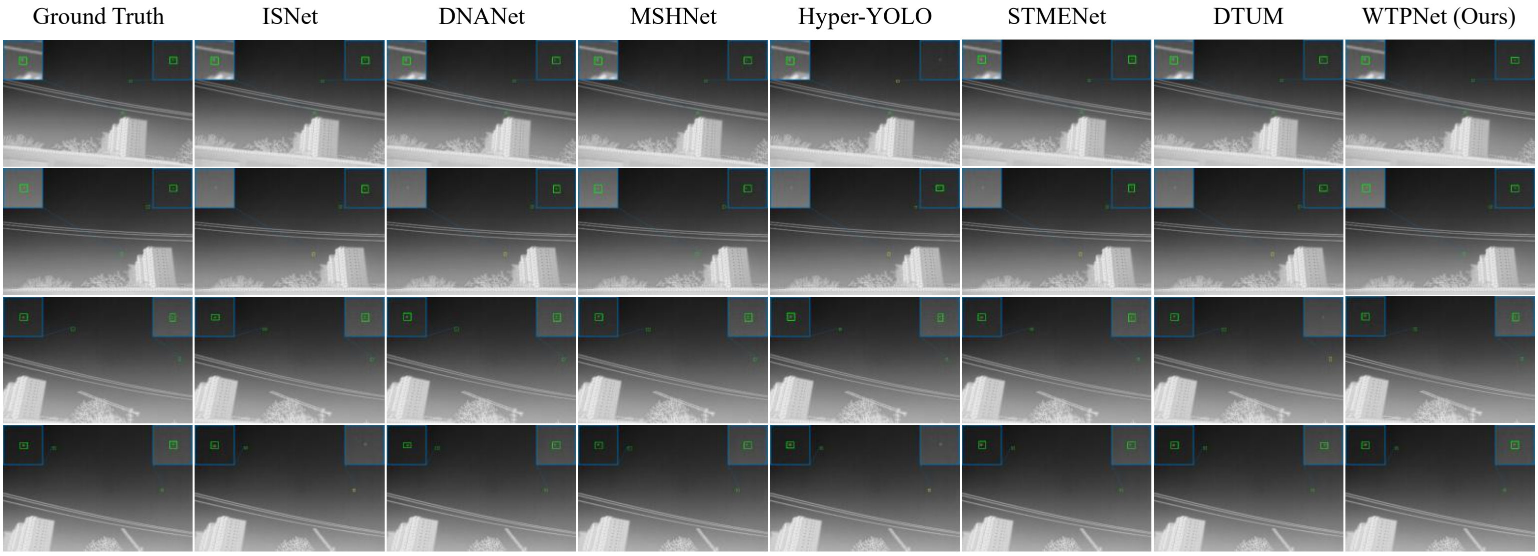


Fig. 8. Visualization comparisons of detection results on IRDST-H. Blue boxes and connecting lines denote the enlarged local regions. Green, yellow, and red boxes indicate correct detections, missed detections, and false alarms, respectively.

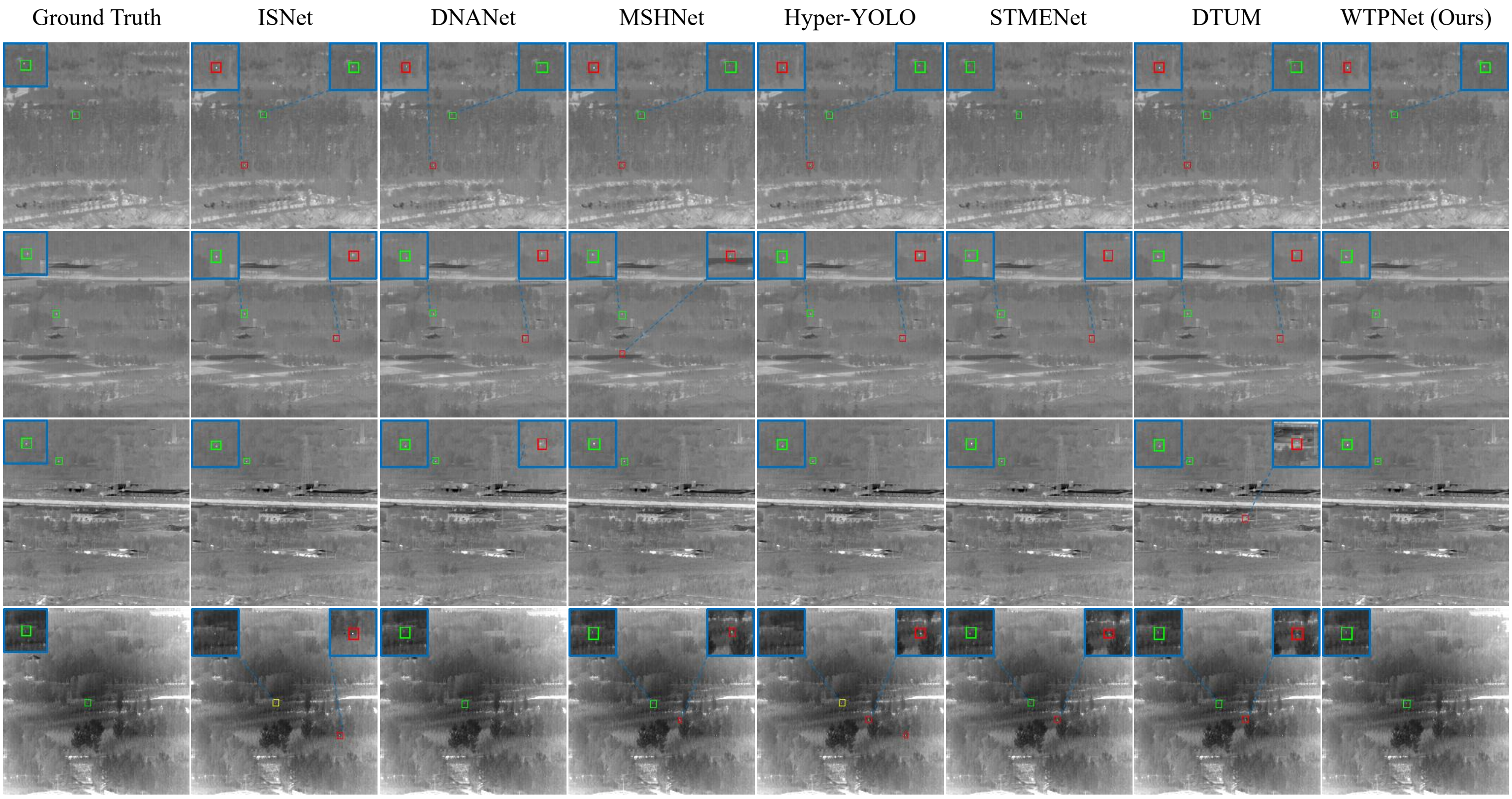


Fig. 9. Visualization comparisons of detection results on DAUB-R. Blue boxes and connecting lines denote the enlarged local regions. Green, yellow, and red boxes indicate correct detections, missed detections, and false alarms, respectively.

4.2.4 Inference Cost Comparisons

Table 3 compares the inference cost and accuracy–efficiency trade-off of different methods on IRDST-H. Here, Params measures the number of trainable parameters, FLOPs reflects the computational complexity of one forward inference, and FPS reports the actual inference throughput. Although WTPNet is not the fastest method due to its five-frame input and temporal feature modeling, it achieves the best detection accuracy with 80.46% F1 and 64.06% AP50. Compared with TDCNet, WTPNet introduces only a small increase in Params and FLOPs, from 25.17M to 25.40M and from 96.92G to 102.35G, respectively, while improving AP50 by 4.96 percentage points. Its FPS decreases moderately from 26.38 to 23.36, indicating that the accuracy gain is obtained with an acceptable runtime cost.

It is also worth noting that WTPNet is faster than several strong single-frame segmentation-based methods, such as DNANet, MSHNet, and RPCANet, despite using five input frames. Compared with most multi-frame methods, WTPNet maintains a comparable inference speed while delivering clearly higher AP50 and F1 scores. As shown in Fig. 10, WTPNet is located at the highest accuracy region of the AP50–FPS plane, forming the accuracy-oriented end of the empirical Pareto envelope. These results indicate that WTPNet prioritizes reliable moving small-target detection without incurring prohibitive computational overhead, achieving a favorable balance between detection accuracy and inference efficiency.

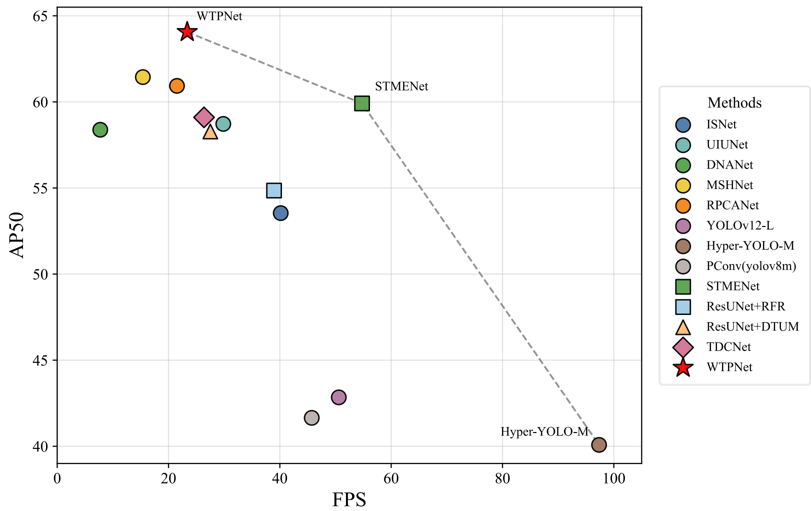


Fig. 10. Accuracy–efficiency trade-off on IRDST-H in terms of AP50 and FPS. The red star denotes WTPNet, and the dashed line indicates the empirical Pareto envelope.

Table 3. Inference cost comparisons on IRDST-H. FPS is measured on an NVIDIA RTX 4090 GPU, and FLOPs are calculated with all input frames resized to 640 × 640. ↑ / ↓ indicate that higher/lower values are better.

Methods	Frames	F1↑	AP50↑	Params(M)↓	FLOPs(G)↓	FPS↑
ISNet	1	72.83	53.54	4.20	70.91	40.16
UIUNet	1	74.78	58.72	55.31	609.41	29.84
DNANet	1	76.14	58.38	7.58	126.63	7.74
MSHNet	1	78.55	<u>61.44</u>	7.11	167.99	15.39
RPCANet	1	77.60	60.93	<u>4.03</u>	389.24	21.52
YOLOv12-L	1	64.63	42.84	27.14	88.90	50.59
Hyper-YOLO-M	1	63.48	40.08	33.59	103.36	**97.35**
PConv(yolov8m)	1	65.12	41.65	23.82	88.62	45.73
SSTNet	5	59.58	35.50	11.95	123.59	20.52
STMENet	5	<u>78.04</u>	59.91	9.85	**41.92**	<u>54.77</u>
ResUNet_RFR	5	73.87	54.85	4.94	<u>54.17</u>	38.96
ResUNet_DTUM	5	76.23	58.28	**3.73**	82.96	27.51
TDCNet	5	77.27	59.10	25.17	96.92	26.38
WTPNet	5	**80.46**	**64.06**	25.40	102.35	23.36

4.3 Ablation Study

4.3.1 Effects of Different Components

To investigate the individual contribution and mutual complementarity of the proposed modules, we conduct an exhaustive component-wise ablation study by enabling WTCP, DPRE, and LCSF under all possible combinations, where the setting without all three modules corresponds to the TDCNet [46] baseline. As reported in Table 4, each individual component brings a clear improvement on IRDST-H. Specifically, WTCP, DPRE, and LCSF improve AP50 from 59.10% to 61.98%, 61.70%, and 61.94%, respectively. These results indicate that wavelet-guided temporal context prior learning, directional photometric residual enhancement, and large-context scale-focused fusion are all beneficial for hard infrared sequences where weak targets are easily disturbed by background clutter.

When multiple components are combined, their complementary roles become clearer. On IRDST-H, WTCP+LCSF achieves 63.53% AP50, and DPRE+LCSF obtains 62.40% AP50, both outperforming their corresponding single-component settings. The full model further increases AP50 to 64.06% and F1 to 80.46%, achieving the best overall performance. This confirms that WTCP provides more discriminative temporal key features, DPRE preserves weak current-frame photometric evidence, and LCSF prevents target responses from being diluted during hierarchical fusion. A similar trend can be observed on DAUB-R. Although some individual modules only bring modest gains due to the relatively high baseline performance, their combination substantially improves AP50. In particular, the full model improves AP50 from 85.85% to 90.11% and recall from 88.71% to 91.12%. The F1 score of the full model is slightly lower than that of the w/o LCSF setting by 0.03 percentage points, but it achieves the highest AP50 and recall, indicating better ranking quality and target retrieval across confidence thresholds.

Overall, the exhaustive ablation results verify the effectiveness of the unified design of WTPNet. The final model consistently achieves the best AP50 on both datasets, demonstrating that temporal prior calibration, shallow photometric response preservation, and scale-focused feature fusion jointly improve the robustness of multi-frame infrared small target detection.

Table 4. Ablation study of different component combinations on IRDST-H and DAUB-R. WTCP, DPRE, and LCSF are progressively enabled to evaluate their individual and complementary effects. The best and second-best results are marked in bold and underlined, respectively.

Settings	WTCP	DPRE	LCSF	IRDST-H				DAUB-R			
				P	R	F1	AP50	P	R	F1	AP50
w/o ALL	-	-	-	73.24	**81.77**	77.27	59.10	97.90	88.71	93.08	85.85
w/ WTCP	√	-	-	78.69	79.54	79.11	61.98	99.48	87.09	92.87	86.95
w/ DPRE	-	√	-	77.43	<u>80.79</u>	79.07	61.70	98.25	89.04	93.42	86.46
w/ LCSF	-	-	√	<u>80.75</u>	77.88	79.29	61.94	**99.74**	86.88	92.87	85.80
w/o LCSF	√	√	-	78.26	78.41	78.34	60.89	<u>99.71</u>	<u>90.76</u>	**95.02**	<u>89.67</u>
w/o DPRE	√	-	√	**81.61**	78.79	<u>80.17</u>	<u>63.53</u>	99.03	87.37	92.84	87.14
w/o WTCP	-	√	√	79.28	79.00	79.14	62.40	99.26	90.35	94.60	88.29
w/ ALL	√	√	√	80.36	80.55	**80.46**	**64.06**	99.22	**91.12**	<u>94.99</u>	**90.11**

4.3.2 Analysis of WTCP

WTCP is designed to refine the context-aware temporal branch by introducing two complementary priors: a DCT-based motion prior from the Q-branch and a Haar wavelet-based context prior from the K-branch. The former captures temporal energy variations related to target motion, while the latter separates low-frequency background structures from high-frequency target-related details before prior fusion and residual injection into the K-branch. To verify this design, we remove each prior source and compare it with the complete WTCP setting. As reported in Table 5, using both priors achieves the best performance, with 80.46% F1 and 64.06% AP50. Removing the motion prior decreases AP50 to 62.85%, while removing the context prior further reduces it to 62.17%. These results indicate that the two priors are complementary rather than interchangeable: the motion prior helps describe frame-wise target-related variations, whereas the wavelet context prior provides background-aware high-frequency discrimination.

Table 5. Ablation study of different prior sources in WTCP on IRDST-H under the full WTPNet setting. The motion prior is derived from temporal DCT, and the context prior is generated from Haar wavelet decomposition. The best result is marked in bold.

Settings	P	R	F1	AP50
w/o motion prior	79.84	78.52	79.17	62.85
w/o context prior	78.27	80.39	79.32	62.17
w/ both (ours)	**80.36**	**80.55**	**80.46**	**64.06**

We further investigate how the generated prior should be injected into the context-aware temporal branch. Since the *K*-branch provides contextual matching cues for subsequent cross-attention, inappropriate prior injection may disturb the original 3D spatio-temporal representation instead of improving it. Table 6 shows that injecting the WTCP prior only into stage 2 achieves the best AP50 of 61.98%, outperforming injection into deeper stages or multiple stages. In contrast, inserting the prior into stages 2, 3, and 4 simultaneously leads to a clear performance drop. This suggests that WTCP is more effective when applied to shallow temporal features, where small-target responses and high-frequency temporal variations are still relatively preserved. As the feature hierarchy becomes deeper, spatial resolution decreases and semantic abstraction increases, directly imposing the prior on these stages may introduce excessive constraint or amplify background-related responses. Therefore, WTCP adopts stage-2 injection as the final setting, allowing the prior to guide early temporal context learning while preserving the stability of deeper spatio-temporal features.

Table 6. Effect of prior injection stages in WTCP on IRDST-H. To isolate the placement effect, only WTCP is enabled while DPRE and LCSF are disabled. The best result is marked in bold.

Inject Stage	P	R	F1	AP50
2,3,4	73.99	**79.87**	76.82	58.30
2,3	<u>78.78</u>	77.98	<u>78.38</u>	<u>60.56</u>
4	78.34	74.93	76.60	57.85
3	**80.66**	75.21	77.84	60.38
2	78.69	<u>79.54</u>	**79.11**	**61.98**

4.3.3 Analysis of DPRE

DPRE is introduced to enhance the spatial appearance branch by preserving weak current-frame photometric responses before they are used as value features in cross-attention. To verify its structural design, we compare two ways of using PConv: directly replacing standard convolutions with PConv, and using PConv as a residual directional enhancement term. As shown in Table 7, the residual enhancement strategy achieves better F1 and AP50, improving AP50 from 59.10% to 61.70%. Although direct replacement obtains higher precision, its recall is lower, indicating that replacing the original convolutional representation may suppress some weak target responses. In contrast, the residual design retains the original CSPDarknet feature pathway while introducing direction-aware photometric compensation, which is more suitable for the value branch that requires stable and reliable appearance evidence.

Table 7. Ablation study of different DPRE settings on IRDST-H. Only DPRE is enabled while WTCP and LCSF are disabled. The best result is marked in bold.

DPRE Settings	P	R	F1	AP50
w/o PConv	73.24	**81.77**	77.27	59.10
Direct PConv Replacement	**79.85**	76.71	78.25	60.52
PConv Residual Enhancement	77.43	80.79	**79.07**	**61.70**

We further study the influence of the residual scaling factor α in Table 8. The best performance is obtained when $\alpha = 0.1$, reaching 79.07% F1 and 61.70% AP50. A smaller value, such as $\alpha = 0.05$, provides weaker enhancement and leads to a lower AP50, while larger values, such as $\alpha = 0.2$ and $\alpha = 0.5$, progressively degrade the performance. This suggests that excessive directional compensation may disturb the original feature distribution and amplify background edge responses. The learnable α setting achieves the highest precision but suffers from a clear recall drop, resulting in lower F1 and AP50. Therefore, a fixed and moderate residual coefficient is more effective for DPRE, as it strengthens weak directional photometric cues without over-modifying the backbone representation.

Table 8. Effect of the residual scaling factor α in DPRE on IRDST-H. Only DPRE is enabled while WTCP and LCSF are disabled.

α	P	R	F1	AP50
0.05	76.74	**80.95**	78.79	61.22
0.1	77.43	80.79	**79.07**	**61.70**
0.2	74.43	80.11	77.16	59.40
0.5	75.99	77.48	76.73	58.26
learnable	**80.39**	73.45	76.76	58.59

4.3.4 Analysis of LCSF

LCSF is designed to address the feature dilution problem in hierarchical fusion, where weak infrared small target responses may be overwhelmed by high-level contextual semantics or background clutter. To study its internal configuration, we first compare different combinations of the large perception kernel K_L and the small aggregation kernel K_S. As shown in Table 9, the configuration with $K_L = 7$ and $K_S = 3$ achieves the best performance, with 79.29% F1 and 61.94% AP50. Larger kernel combinations, such as $(15,5)$ and $(13,5)$, lead to obvious performance degradation. This suggests that overly broad perception and large local aggregation may introduce excessive background responses, which is unfavorable for compact infrared targets. In contrast, the $(7,3)$ setting provides a better balance: the large-kernel branch still captures sufficient contextual information for clutter suppression, while the small-kernel aggregation preserves local target details.

Table 9. Effect of kernel configurations in LCSF on IRDST-H. K_L and K_S denote the large perception kernel size and the small aggregation kernel size, respectively. Only LCSF is enabled while WTCP and DPRE are disabled.

K_L	K_S	P	R	F1	AP50
15	5	79.28	73.82	76.45	57.70
13	5	73.39	74.64	74.01	54.07
11	3	77.37	**78.23**	77.80	60.19
9	3	**80.98**	76.51	<u>78.68</u>	<u>61.06</u>
7	3	<u>80.75</u>	<u>77.88</u>	**79.29**	**61.94**

We further compare LCSF with several representative feature fusion strategies in Table 10. Compared with Plain FPN, BiFPN, and ASFF, LCSF achieves the highest F1 and AP50. Plain FPN provides a simple top-down fusion pathway but lacks adaptive scale focusing. BiFPN introduces bidirectional feature interaction, yet its repeated weighted fusion may not be optimal for extremely small and weak targets. ASFF obtains relatively high recall but lower precision and AP50, indicating that adaptive scale fusion alone is insufficient to suppress background interference. By contrast, LCSF introduces large-context-guided local aggregation after feature fusion, allowing the network to perceive surrounding clutter while maintaining compact target responses. The feature heatmaps in Fig. 11 further support this observation. Plain FPN, BiFPN, and ASFF tend to produce broad or clutter-sensitive activations, especially around strong background structures. In comparison, LCSF generates more concentrated responses around true target locations while suppressing large-area background activations. These results indicate that LCSF improves multi-scale fusion not merely by combining features, but by guiding the fused representation to focus on target-relevant local regions under broader contextual awareness.

Table 10. Comparison with different feature fusion strategies on IRDST-H. Different fusion modules are evaluated under the same setting, with WTCP and DPRE disabled.

Feature Fusion	P	R	F1	AP50
Plain FPN	<u>78.71</u>	<u>77.12</u>	<u>77.91</u>	<u>60.37</u>
BiFPN	77.25	73.91	75.45	56.13
ASFF	76.95	76.00	76.47	58.21
LCSF	**80.75**	**77.88**	**79.29**	**61.94**

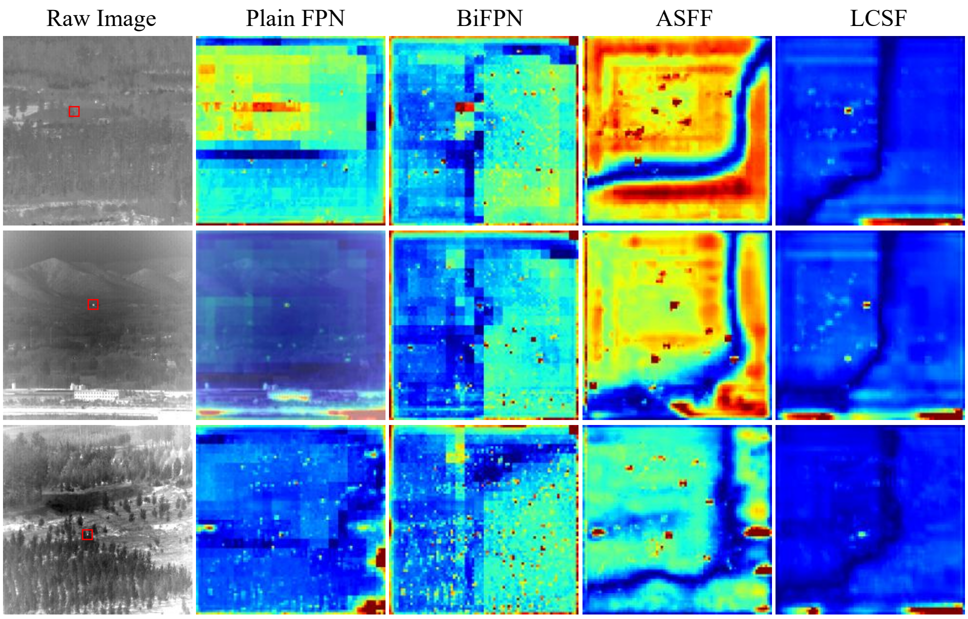


Fig. 11. Feature response visualization of different feature fusion strategies on IRDST-H. From left to right: input image, Plain FPN, BiFPN, ASFF, and LCSF.

4.3.5 Effect of Input Frame Number

We further investigate the influence of the input frame number T on detection performance, as shown in Fig. 12. The results indicate that increasing the temporal length from 3 to 5 consistently improves AP50 on both datasets, demonstrating that a moderate number of adjacent frames provides useful temporal cues for moving infrared small target detection. The best performance is achieved when $T = 5$, with AP50 values of 64.06% on IRDST-H and 90.11% on DAUB-R. However, further increasing the frame number does not bring additional gains. When T is increased to 6 or 7, AP50 decreases on both datasets, especially on DAUB-R. This suggests that excessively long frame sequences may introduce redundant temporal information, larger target displacement, and background variation, which can weaken the effectiveness of temporal context modeling. Therefore, we set $T = 5$ in all experiments to balance temporal information utilization and interference suppression.

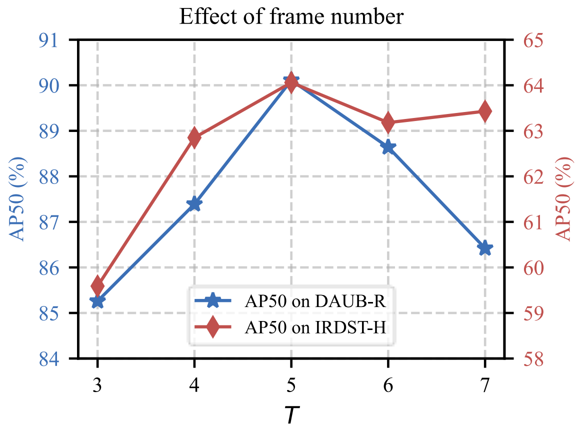


Fig. 12. Impact of the input frame number on AP50 over IRDST-H and DAUB-R.

5. Conclusion

In this paper, we proposed WTPNet, a wavelet-guided temporal photometric network for multi-frame infrared small target detection. Instead of directly stacking consecutive frames, WTPNet decouples multi-frame representation learning into motion-sensitive temporal modeling, context-aware temporal prior learning, and current-frame spatial appearance preservation. Specifically, WTCP introduces DCT- and Haar wavelet-guided priors to enhance target-related temporal context, DPRE strengthens weak shallow photometric responses through residual directional enhancement, and LCSF performs large-context-guided scale-focused fusion to alleviate the dilution of tiny target responses. Experiments on IRDST-H and DAUB-R demonstrate that WTPNet achieves superior performance over representative single-frame and multi-frame methods, with consistent improvements in quantitative metrics, PR curves, visual comparisons, and ablation studies.

Despite its effectiveness, WTPNet still has several limitations. The three-stream cross-attention design improves discriminative representation but is not sufficiently lightweight for resource-constrained onboard or edge deployment. In addition, the current framework adopts a fixed input frame number and may still be affected by large target displacement, abrupt background changes, or long-term occlusion. Future work will focus on hardware-aware lightweight optimization, including pruning, quantization, and deployment-oriented acceleration. Another important direction is to develop adaptive temporal selection and alignment strategies, as well as more fine-grained evaluations for different target categories, motion patterns, and signal-to-clutter conditions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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